

E-Commerce Business Analysis Report

Project Objective

Customer churn represents a significant risk to e-commerce businesses, as it directly impacts revenue and long-term customer value. This project analyzes customer behavior data to identify the key drivers of churn and generate actionable insights to improve customer retention.

This project uses an e-commerce customer dataset to:

- Examine the customer data to correlate different indicators of churn
- Examine how certain categories correlate with lifetime value, including recency (Days_Since_Last_Purchase), discount usage (Discount_Usage_Rate), and engagement (Pages_Per_Session)
- Analyzing the impact of cart abandonment on purchasing behavior
- Developing a logistic regression model to assess the relative importance of different churn drivers

The objective of this project is to translate behavioral patterns into practical business recommendations that enable more effective retention strategies and reduce customer attrition.

Key Business Questions

This analysis is designed to address the following business questions:

1. What behavioral factors (e.g., recency, activity, engagement) are the primary drivers of customer churn?
2. How does churn vary across different customer segments?
3. What is the overall churn rate, and how does it change based on customer behavior?
4. Which factors are most strongly associated with higher customer lifetime value?
5. How does cart abandonment impact purchasing behavior and revenue generation?
6. Are missing or incomplete behavioral signals indicative of churn risk?
7. Which customer segments are most at risk of churning and should be targeted for retention efforts?
8. What actionable strategies can be implemented to reduce churn and improve customer retention?

Analytical Stack

This project follows a structured data analytics workflow using three tools:

Python (Data Preprocessing, Exploration, Logistic Model Creation, and Analysis)

Python was used to perform data exploration, assess data quality, analyze customer behavior, and build a predictive churn model. The focus was on understanding relationships between behavioral variables and key outcomes such as churn and customer lifetime value.

1. Data Exploration and Quality Assessment

Initial data exploration was conducted to understand the structure, distributions, and completeness of the dataset.

Key steps included:

- Evaluating missing values across all variables
- Creating missing data indicators for key behavioral features
- Identifying potential data quality issues (e.g., inconsistent or extreme values)
- Verifying data types to ensure proper analysis

This step ensured the dataset was reliable and suitable for further analysis while also identifying whether missing data carried any meaningful signal.

2. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand overall patterns and distributions within the data.

This included:

- Analyzing distributions of key variables such as recency, engagement, and activity
- Identifying trends in customer behavior
- Examining summary statistics (mean, median, quartiles)
- Detecting outliers and unusual patterns

EDA provided a foundational understanding of how customers interact with the platform and highlighted variables likely to influence churn and value.

3. Logistic Regression Model

A logistic regression model was developed to evaluate the relationship between customer behavior and churn probability.

Key aspects:

- The model used behavioral variables (e.g., recency, engagement, activity, cart abandonment) as predictors
- Coefficients were analyzed to determine the direction and strength of each variable's impact on churn
- Model performance was evaluated using metrics such as precision, recall, and accuracy

The model achieved **77% accuracy**, but recall for churned customers was relatively low at **27%**, indicating limited effectiveness in identifying at-risk users. This is likely due to class imbalance, where non-churned customers significantly outnumber churned customers.

Despite this limitation, the model provided valuable insights into key behavioral drivers:

- **Higher cart abandonment rates** are associated with increased likelihood of churn
- **Longer periods of inactivity (recency)** significantly increase churn risk
- **Higher engagement levels** (e.g., more pages per session) are associated with reduced churn

This model helped identify the most influential drivers of churn and supported findings from exploratory analysis.

4. Univariate Analysis

Univariate analysis was conducted to examine individual variables independently.

This included:

- Reviewing summary statistics for key variables
- Understanding the distribution and spread of features such as:
 - Days Since Last Purchase
 - Pages Per Session
 - Session Duration
 - Cart Abandonment Rate

Univariate analysis of key behavioral metrics revealed several important patterns. Cart abandonment rates are generally high across the dataset, indicating potential friction in the purchasing process. The distribution of recency shows a long tail of inactive users, suggesting a substantial portion of customers are at risk of disengagement. Additionally, engagement metrics (such as pages per session and session duration) display significant variation, highlighting clear differences in user activity levels.

These patterns align with findings from the churn model, reinforcing recency and purchase friction as primary drivers of customer attrition, while higher engagement is associated with stronger customer retention.

5. Bivariate Analysis

Bivariate analysis was performed to evaluate relationships between variables and key outcomes such as churn and lifetime value.

Key analyses included:

- Recency vs churn
- Engagement vs lifetime value
- Cart abandonment vs total purchases
- Discount usage vs lifetime value

Bivariate analysis was conducted to evaluate relationships between key behavioral variables and business outcomes such as churn and customer lifetime value.

The analysis revealed that **customer engagement** is a primary driver of lifetime value, with more engaged users generating higher overall revenue. In contrast, **cart abandonment** shows a clear negative relationship with purchasing behavior, where higher abandonment rates are associated with lower purchase frequency.

Recency (days since last purchase) was found to strongly influence churn risk, with less active customers significantly more likely to churn. However, recency does not appear to have a direct relationship with historical lifetime value, indicating it is more predictive of future behavior than past revenue.

Finally, **discount usage** showed limited impact on long-term customer value, suggesting that promotional strategies alone may not drive sustained customer retention.

SQL (Data Transformation, Aggregation, and Analytical Querying)

SQL was used to perform structured data analysis, aggregation, and validation of customer churning behaviors. After initial data preparation in Python, the cleaned dataset was loaded into a relational database to enable efficient querying and scalable analysis.

1. Data Ingestion & Schema Design

A two-step loading process was implemented to ensure data quality and type consistency:

- Imported raw data into a **staging table** with flexible data types (NVARCHAR)
- Applied **safe type conversions (TRY_CONVERT)** to load data into a structured final table
- Defined appropriate data types (INT, DECIMAL, BIT) for accurate calculations and performance

This approach minimized data loss and ensured robustness during ingestion.

2. Core Metric Calculation

SQL was used to aggregate customer behavior data, calculate business metrics, and identify high-risk customer segments. Queries focused on evaluating relationships between customer behavior, churn risk, engagement, and lifetime value.

Example techniques used:

- `AVG(CAST(Churned AS FLOAT)) AS Churn_Rate` for churn rate
- `CASE WHEN Days_Since_Last_Purchase <= 10 THEN '0-10 Days'` for segmenting days of recency
- `COUNT(*) AS Customers` for total customers
- Filtering logic to identify at-risk customers

3. Churn by Recency

Customers were segmented based on the number of days since their last purchase to evaluate how inactivity impacts churn risk.

The analysis revealed a strong relationship between recency and churn, where customers with longer periods of inactivity exhibited significantly higher churn rates. This reinforces recency as one of the strongest behavioral indicators of customer attrition.

4. Engagement vs Lifetime Value

Customers were grouped by engagement level using pages per session to evaluate how platform interaction influences customer value.

Results showed that highly engaged customers generated the highest average lifetime value, despite representing a smaller portion of the customer base. This suggests that increasing customer engagement may lead to stronger long-term revenue generation.

5. Cart Abandonment vs Purchase

Customers were categorized by cart abandonment rate to analyze its relationship with purchasing behavior and customer value.

The analysis identified a negative relationship between cart abandonment and purchasing activity. Customers with higher abandonment rates tend to make fewer purchases and generate lower lifetime value, indicating potential friction within the purchasing funnel.

6. Churn By Engagement

Login frequency was used to evaluate how customer activity levels impact churn.

Results showed that customers with lower activity levels experience significantly higher churn rates, while highly active users demonstrate stronger retention. This highlights customer engagement as an important factor in long-term retention.

7. At-Risk Customers

SQL filtering logic was used to identify customers with high churn risk based on multiple behavioral conditions, including:

- `Days_Since_Last_Purchase > 30`

- Login_Frequency < 10
- Cart_Abandonment_Rate > 60

This query enabled the identification of customers who may benefit from proactive retention strategies and targeted re-engagement campaigns.

8. Missing Data vs Churn

Missing behavioral indicators were analyzed to determine whether incomplete customer activity data was associated with churn risk.

The analysis found minimal differences in churn rates between customers with and without missing behavioral data, suggesting that missingness alone is not a strong predictor of churn within this dataset.

Overall, SQL analysis reinforced findings from Python-based exploratory analysis and helped translate behavioral trends into actionable business metrics and customer segments.

Power BI (Data Visualization & Interactive Dashboarding)

Power BI was used to build an interactive dashboard designed to communicate customer churn insights, behavioral patterns, and retention risk in a clear, business-focused format.

1. Data Integration

- Imported the cleaned dataset from Python
- Validated row counts and key metrics to ensure consistency across tools
- Structured the dataset for efficient visualization , filtering, and segmentation

2. Data Modeling & Measures

Custom measures were created using DAX to support analysis:

- Activity Level
- Average Lifetime Value (AVG LTV)
- Churn Rate
- Engagement Level

- Recency Group
- Total Customers

3. Dashboard Design

The dashboard was designed to provide both high-level KPI monitoring and deeper behavioral insights into customer churn risk:

- Total Customers
- Churn Rate
- Average Lifetime Value
- Churn Rate by Customer Recency
- Churn Rate by Customer Activity Level
- Customer Value by Engagement Level
- Purchases by Cart Abandonment
- Churn Rate by Missing Behavioral Data
- Churn Rate by Lifetime Value Segment

The dashboard visualizations reinforce several key findings from the analysis:

- Churn risk increases significantly as customer inactivity rises
- Lower login frequency is associated with reduced customer retention
- Higher engagement levels correlate with stronger customer value
- Increased cart abandonment is linked to lower purchasing activity

Together, these visualizations provide a centralized view of the behavioral patterns most strongly associated with customer attrition.

4. Interactivity & User Experience

- Implemented slicers (country, gender, signup quarter, recency group) for dynamic filtering and exploratory analysis
- Designed visuals to support intuitive comparison between customer segments and behavioral trends

5. Key Value of Power BI in This Project

Power BI enabled:

- Clear communication of customer churn behaviors and retention trends
- Interactive exploration of customer segments and behavioral metrics
- Centralized visualization of recency, engagement, activity, and churn risk indicators

Expected Business Impact

The findings from this analysis provide actionable insights that can help e-commerce businesses improve customer retention, reduce churn, and increase long-term customer value.

Key expected business impacts include:

Improved Customer Retention

Identifying high-risk customers based on inactivity, low engagement, and cart abandonment enables businesses to implement targeted retention and re-engagement strategies before customers churn.

Higher Customer Lifetime Value

The analysis highlights customer engagement as a strong driver of lifetime value, suggesting that improving user interaction and platform engagement may increase long-term revenue generation.

Reduced Purchase Friction

High cart abandonment rates indicate potential friction within the purchasing process. Addressing checkout inefficiencies and improving the user experience may lead to increased conversions and purchase frequency.

More Effective Customer Segmentation

Behavioral segmentation allows businesses to better understand customer groups and tailor marketing, retention, and engagement strategies accordingly.

Data Driven Decision Making

The Power BI dashboard provides stakeholders with an interactive tool for monitoring churn trends, customer activity, and engagement patterns, supporting faster and more informed business decisions.

Overall, this project demonstrates how customer behavior analytics can be leveraged to proactively identify churn risk and support retention-focused business strategies.